

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

6th Semester of B. Pharm. Examination

University Theory Examination November 2016

PH 313 Biopharmaceutics and Pharmacokinetics

Date: 10/11/2016, Thursday Time: 10:00 A.M TO 1:00 P.M Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following.

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| A | What is biopharmaceutics? Explain its role in formulation development. Enlist various barriers to drug absorption. Describe passive diffusion of drug. | 10 |
| B | Explain drug transport. Enlist the various mechanism of transport of drug across biological barriers. Discuss any one in detail. | 10 |
| C | Write a note on factors affecting metabolism of drug. Discuss the importance of protein binding of drugs. | 10 |

Q 2 Attempt any TWO of the following.

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| A | What is Vd? Describe the importance of volume of distribution in detail. Discuss tissue binding of drugs in detail. | 10 |
| B | What is BCS classification? How does it affects development of dosage form? Discuss the significance of BCS system. | 10 |
| C | Classify the methods for bioavailability measurement. Discuss the method based on plasma data. Describe Latin Square cross over design for bioequivalence study. | 10 |

SECTION- II

Q 3 Attempt any TWO of the following

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| A | What is Pharmacokinetic study? Explain how plasma drug concentration can be measured? What is the importance of it? | 10 |
| B | Explain Wagner Nelson and Loo Riegelman method in brief. | 10 |
| C | Write a shortnote on "Renal Clearance". | 10 |

(P.T.O.)

Q 4	Attempt any <u>TWO</u> of the following	
A	Explain in brief about nonlinear one compartment model after I.V. drug administration.	10
B	a) Write the concept and significance of Bioavailability and Bioequivalence.	05
	b) Explain the following terms in brief. T _{max} , C _{max} , AUC, IVIVC, Biosimilar drugs	05
C	Write a shortnote : Factors affecting drug absorption.	10